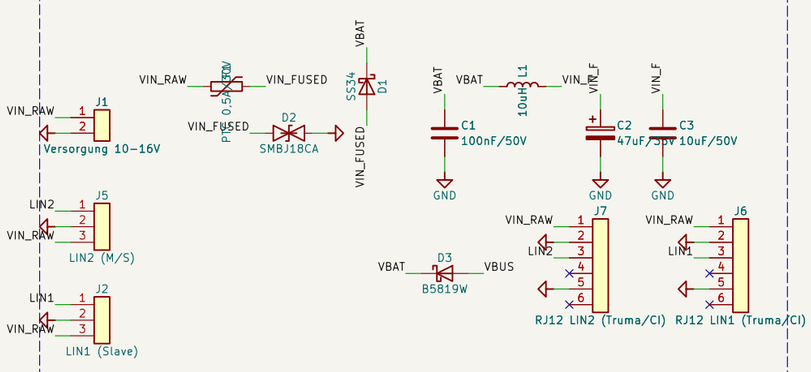
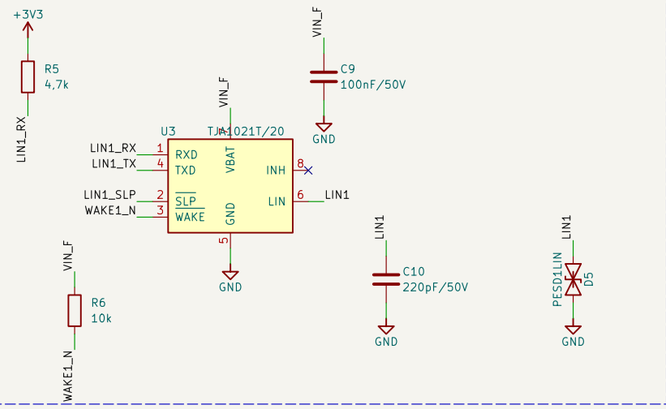
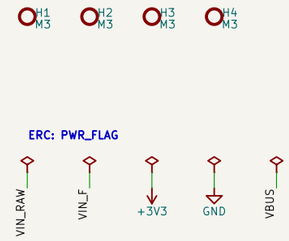
Einspeisung: Sicherung, Verpolschutz, TVS, Pi-Filter

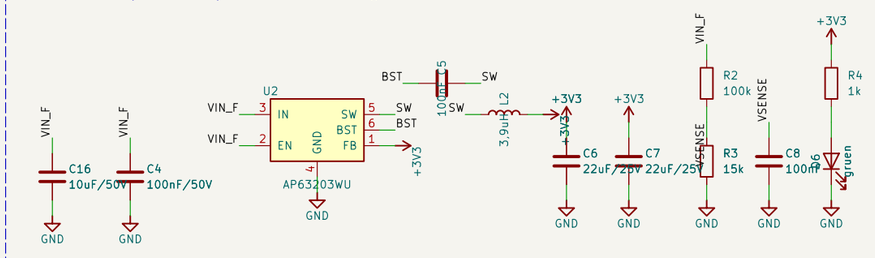


LIN1 – TJA1021, fest Slave

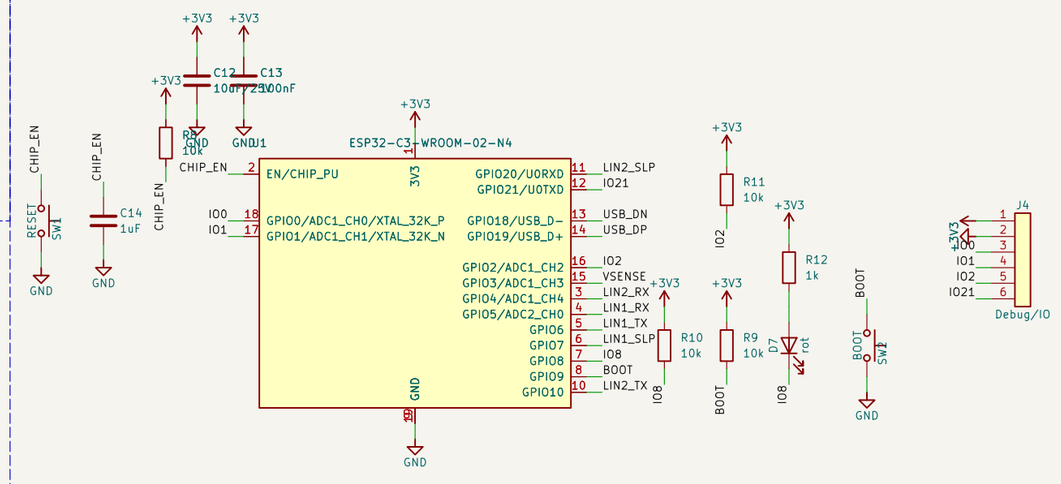




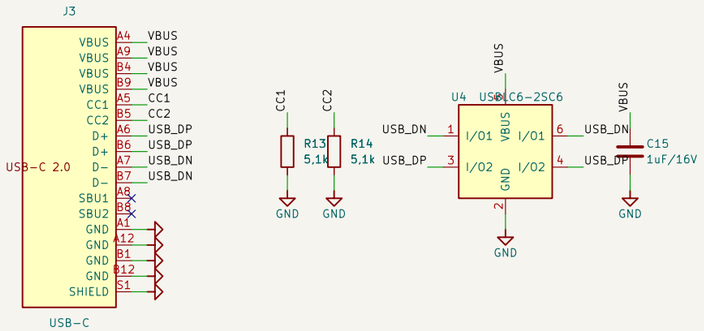
Buck 3,3 V (AP63203) + VIN-Messung + Power-LED



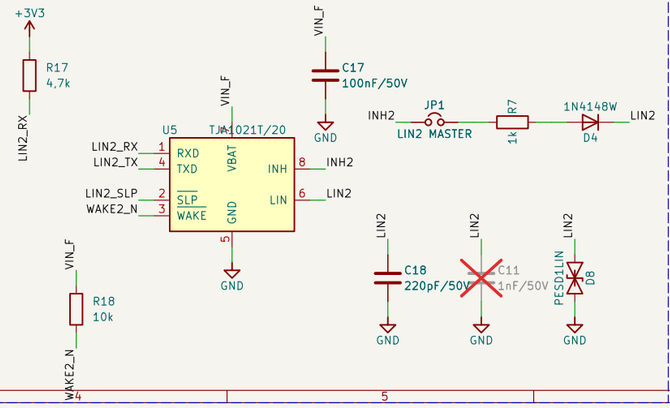
ESP32-C3-WROOM-02



USB-C (Flashen / Debug, USB-Serial/JTAG)



LIN2 – TJA1021, Master/Slave per Steckbruecke JP1



Hinweise:

- Eingang 10–16 V (Labor/Wohnmobil). TVS SMBJ18CA, Verpolung: TVS leitet -> PTC F1 loest aus, SS34 sperrt.
- Pi-Filter C1 / L1 10uH / C2 47uF Elko (Daempfung) + C3 10uF gegen Leitungsstoerungen des Buck (1,1 MHz).
- Buck dauerhaft an (EN an VIN), kein Schlafmodus. LIN1 fest Slave. LIN2: Steckbruecke JP1 gesetzt = Master (1k + Diode an INH2).
- LIN2 Master optional: C11 1nF bestuecken und C18 entfernen (220pF funktioniert in der Praxis meist auch).
- LIN1: RX=GPIO5 TX=GPIO6 SLP=GPIO7 (UART1). LIN2: RX=GPIO4 TX=GPIO10 SLP=GPIO20 (UART0 umgemappt). VSENSE=GPIO3, VIN=VSENSE x 7,67.
- RJ12 J6 (LIN1) / J7 (LIN2) parallel zu den Schraubklemmen, Truma/CI-Bus-Belegung: 1=+12V (VIN_RAW), 2=GND, 3=LIN, 4=frei, 5=GND, 6=frei.
- USB-C nur zum Flashen/Debuggen (USB-Serial/JTAG an GPIO18/19). Status-LED an GPIO8 (Low = an).

Rev 4.0: Schaltung unveraendert ggue. 3.0; Layout 100x80 mm, JLCDFM-bereinigt
2x LIN: LIN1 Slave, LIN2 Master/Slave per Steckbruecke JP1

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Title: ESP32-C3 2-Kanal-LIN-Knoten (2x TJA1021), 10–16 V

Size: A3	Date: 2026-10-31	Rev: 4.0
KiCad E.D.A. 10.0.8		Id: 1/1